EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2018
General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME					
CIVICS GROUP	1	7	–		

INDEX NUMBER		

CHEMISTRY

8873/02

Paper 2 Structured Questions

13 September 2018

2 hours

Candidates answer **Section A** and **Section B** on the Question Paper

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on the work you hand in.

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question paper.

Section B
Answer **one** question in the spaces provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question..

For Examiner's Use	
Section A (60 marks)	
1	
2	
3	
4	
5	
Section B (20 marks)	
Total	

This document consists of 22 printed pages.

Section A

Answer **all** the questions in this section.

For Examiner's Use

1 Sulfur, atomic number 16, is found within the Earth's crust. Sulfur is released into the atmosphere at times of volcanic activity.

A sample of sulfur from a volcano was analysed to give the following composition of isotopes.

isotope	**abundance (%)**
^{32}S	95.02
^{33}S	0.76
^{34}S	4.22

(a) Define the term relative atomic mass.

....

....

....

....[1]

(b) Calculate the relative atomic mass of the sample of sulfur.

[1]

(c) John Dalton, an early 19th century scientist, believed that elements were made up of tiny particles called atoms which could not be divided. Nowadays, chemists know of the existence of sub-atomic particles in atoms and in ions.

Complete the table to show the number of sub-atomuration of the ^{33}S atom and $^{34}S^{2-}$ ion.

isotopic species	**number of protons**	**number of neutrons**	**number of electrons**	**electronic configuration**
^{33}S				$1s^2$ ....
$^{34}S^{2-}$				$1s^2$ ....

[4]

For Examiner's Use

(d) Solid sulfur exists as a lattice of S_8 molecules. Each S_8 molecule is a ring of eight atoms.

How many atoms of sulfur are there in 0.0120 mol of S_8 molecules?

[1]

(e) The only intermolecular forces in solid sulfur are instantaneous dipole-induced dipole.

(i) Describe how instantaneous dipole-induced dipole arise.

....

....

....

....[2]

(ii) Suggest why there are no other intermolecular forces in solid sulfur.

....

....

....[1]

(f) Sulfur hexafluoride, SF_6, exists as non-polar covalent molecules with an octahedral shape.

Explain why a molecule of SF_6 has an octahedral shape.

....

....

....

....

....[2]

[Total: 12]

For Examiner's Use

2 Iodine, I_2, is a grey-black solid that is not very soluble in water.

Equilibrium 1 is set up with the equilibrium position well to the left.

$$I_2(s) \rightleftharpoons I_2(aq) \qquad \textbf{equilibrium 1}$$

Solid iodine is much more soluble in an aqueous solution of potassium iodide, KI(aq), than in water.

Equilibrium 2 is set up.

$$I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq) \qquad \textbf{equilibrium 2}$$

(a) Suggest why I_2 is **not** very soluble in water.

....

....[1]

(b) **(i)** Write an expression for the equilibrium constant for this reaction, K_c, stating its units.

[2]

When 2.54 g of solid iodine is dissolved in 100 cm^3 of 1.00 mol dm^{-3} KI, and allowed to reach equilibrium, the resulting $[I_3^-(aq)]$ = 9.98×10^{-2} mol dm^{-3}.

(ii) Calculate the equilibrium values of $[I_2(aq)]$ and $[I^-(aq)]$ in this solution and hence calculate a value for K_c.

$[I_2(aq)]$ = ...

$[I^-(aq)]$ = ...

K_c = ...[3]

For Examiner's Use

(c) The student adds an excess of aqueous silver nitrate, $AgNO_3(aq)$, to the equilibrium mixture forming a yellow AgI precipitate.

State what other observation will be seen and explain the observations in terms of both **equilibrium 1** and **equilibrium 2** and any species formed.

....

....

....

....

....[2]

(d) Two redox reactions of iodine are described below.

Reaction 1: Iodine is reacted with oxygen to form a compound with a molar mass of 333.8 g mol^{-1}.

Reaction 2: Under alkaline condition, iodine disproportionates to form iodide ions and iodate(V), IO_3^- ions.

Construct equations for these **two** reactions.

Reaction 1:

....

Reaction 2:

....[2]

[Total: 10]

3 Table 3.1 gives some data on four fuel sources: methanol, ethanol, hydrogen and octane. Octane can serve as a rough approximation of petrol.

For Examiner's Use

Table 3.1

compound	**formula**	**molar mass / g mol^{-1}**	**density / g cm^{-3}**	**$\Delta H_c^{\ominus}$ (298 K) / kJ mol^{-1}**	**$\Delta H_f^{\ominus}$ (298 K) / kJ mol^{-1}**
methanol	CH_3OH	32.0	0.793[a]	–726.0	–239.1
ethanol	C_2H_5OH	....	0.789[a]	–1367.3	–277.1
liquid hydrogen	H_2	2.0	0.0711[b]		
octane	C_8H_{18}	....	0.703[a]		–250.0

[a] At 298 K and 1 bar pressure
[b] At 20 K and 1 bar pressure

(a) Insert the missing molar mass values in the Table 3.1. [1]

(b) Calculate the density of **gaseous** hydrogen at 298 K and 1 bar pressure.
Assume 1 mol of any gas occupies 24 dm^3 at 298 K and 1 bar pressure.
Give your answer in g cm^{-3}.

density = ... g cm^{-3} [1]

(c) What is the value of the standard enthalpy of formation of hydrogen **gas**, H_2?

....[1]

(d) Use the information in Table 3.2 to give the value of the standard enthalpy of combustion of hydrogen at 298 K.

Table 3.2

compound	**$\Delta H_f^{\ominus}$ (298 K) / kJ mol^{-1}**
water	–285.8
carbon dioxide	–393.5

$\Delta H_c^{\ominus}$ of hydrogen at 298 K = ...kJ mol^{-1} [1]

For Examiner's Use

(e) Write down the chemical equation that represents the *standard enthalpy of combustion* of octane. Include state symbols.

....[1]

(f) Use the enthalpy of formation data in Table 3.1 and Table 3.2 and the equation in **(e)** to calculate the standard enthalpy of combustion, $\Delta H_c^\ominus$, of octane.

[2]

(g) An important property of a fuel, especially when the fuel has to be lifted (such as in aviation), is the energy released on combustion *per gram* of fuel.

Calculate the enthalpy change of combustion per gram of fuel at 1 bar pressure and 298 K for methanol and hydrogen gas.

(i) methanol

$\Delta H_c^\ominus$ = ... kJ g^{-1} [1]

(ii) hydrogen gas

$\Delta H_c^\ominus$ = ... kJ g^{-1} [1]

For Examiner's Use

(h) Another important characteristic of a fuel, especially when there is a fuel tank of limited size, is the energy released on combustion *per cm³* of fuel.

Calculate the enthalpy change of combustion per cm^3 of fuel for ethanol and octane.

(i) ethanol

ΔH_c° = ... kJ cm^{-3} [1]

(ii) octane

ΔH_c° = ... kJ cm^{-3} [1]

(i) Explain why, given the data in the question, it is not strictly possible to make a fair comparison of the energy released per cm^3 of liquid hydrogen with the other fuels.

....

....[1]

[Total: 12]

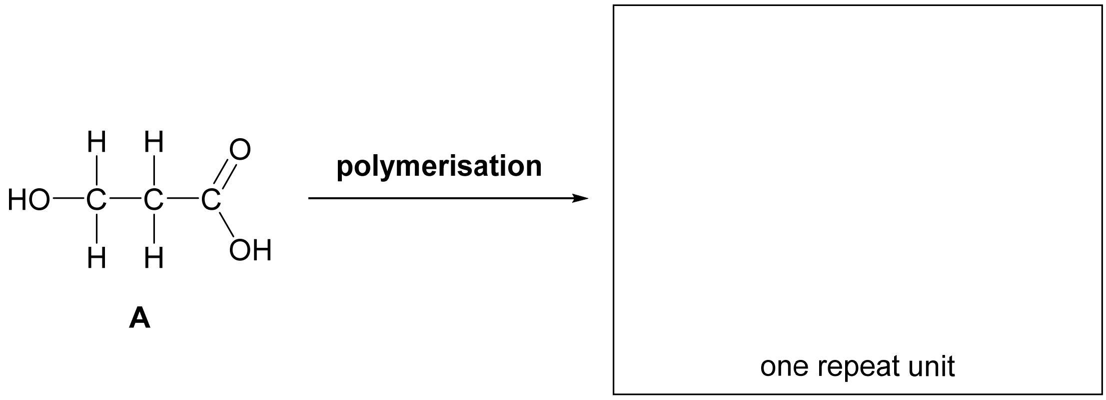
4 **(a)** 3-Hydroxypropanoic acid, $HOCH_2CH_2COOH$, can be produced microbiologically from sugars in corn. $HOCH_2CH_2COOH$ can be used as a 'green' starting material for the synthesis of many organic compounds including some important polymers.

Three synthetic routes are shown below for converting $HOCH_2CH_2COOH$ (**A**) into different polymers.

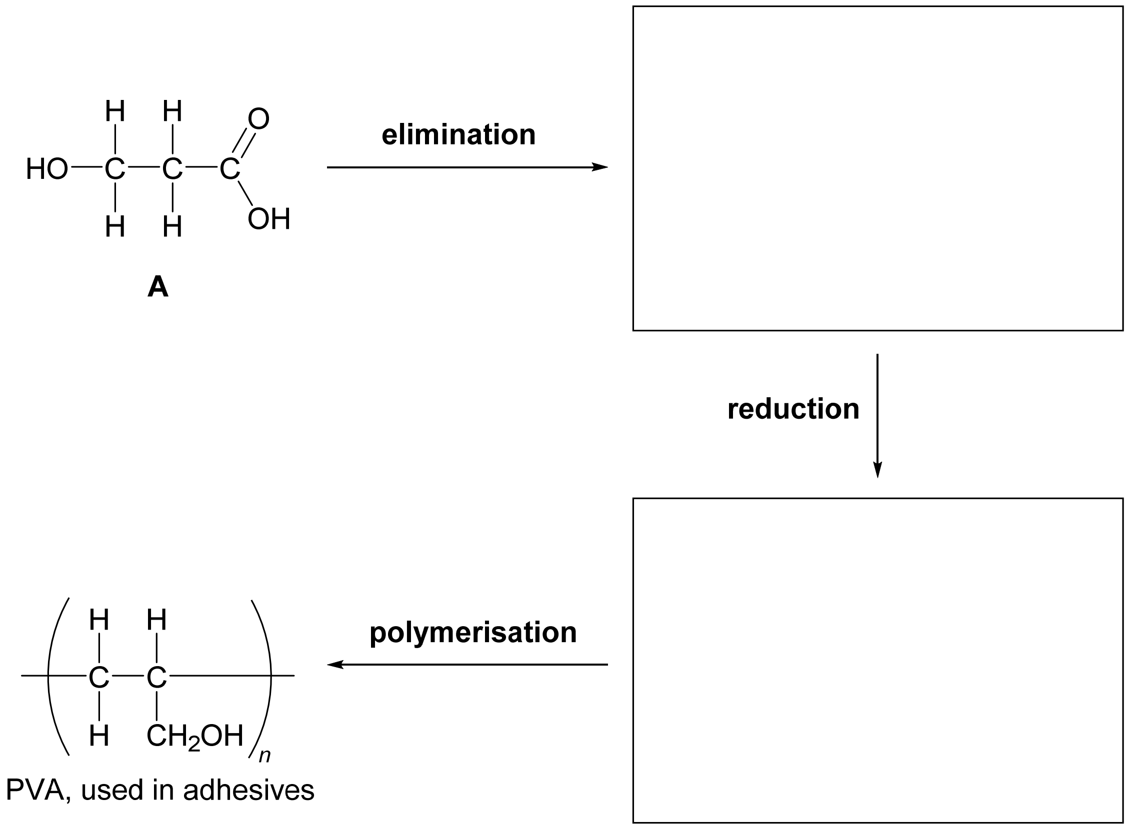
The names of the processes for each synthetic step are given.

(i) In the boxes below, give the structures of the organic compounds formed.

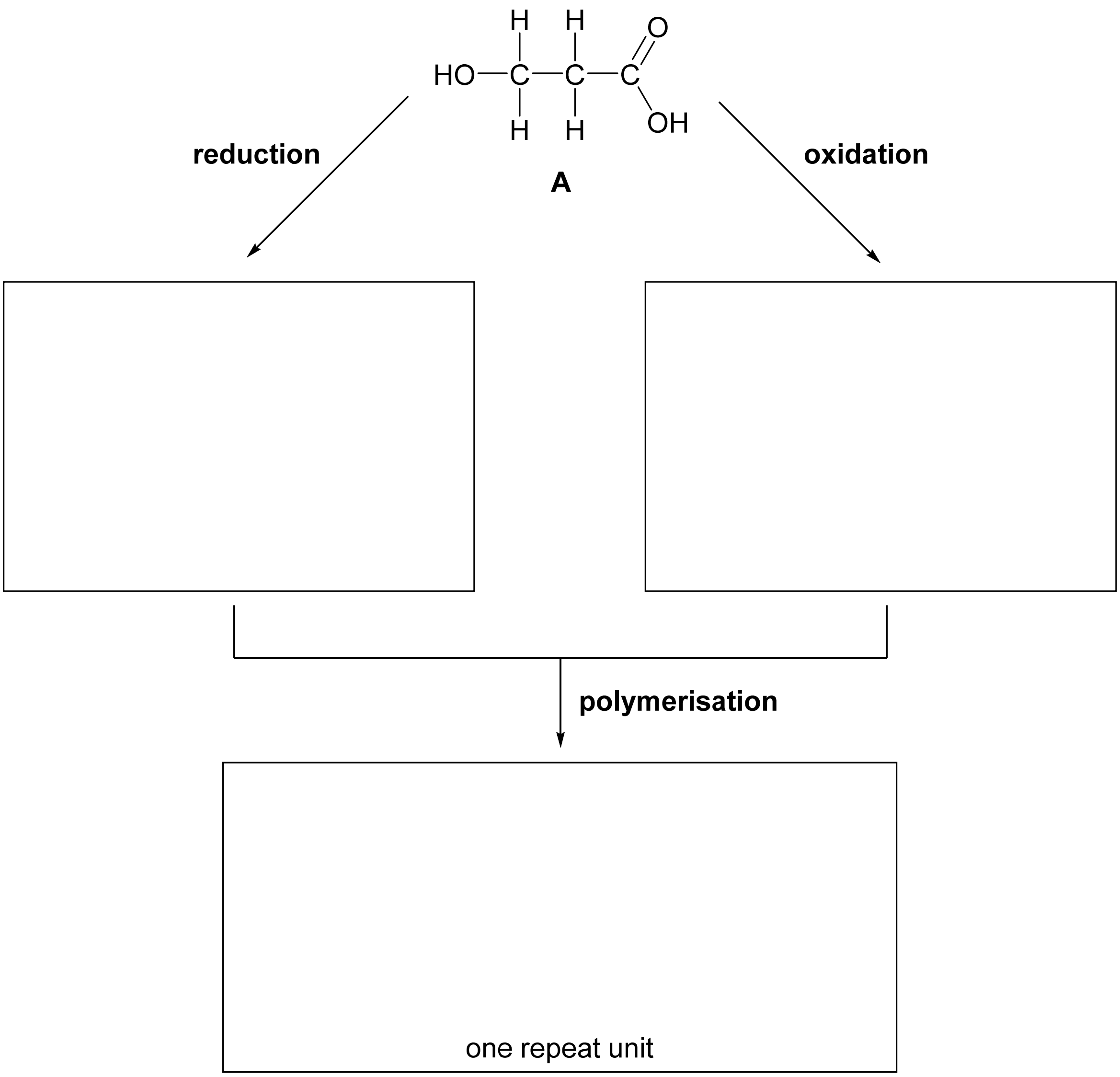
<u>**Synthesis 1**</u>



<u>**Synthesis 2**</u>



Synthesis 3



[6]

(ii) State the type of polymerisation taking place in each synthetic route.

Synthesis 1: .... polymerisation

Synthesis 2: .... polymerisation

Synthesis 3: .... polymerisation [1]

(iii) Name the reagent and condition used in the **reduction** process in **synthesis 3**

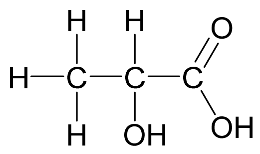
....[1]

(iii) Name the reagent and condition used in the **oxidation** process in **synthesis 3**

....[1]

For Examiner's Use

(b) 2-Hydroxypropanoic acid, also known as lactic acid, has structure shown below.



(i) When heated strongly, lactic acid forms a cyclic 'diester'.

The diester has the molecular formula, $C_6H_8O_4$.

Draw the structure of the cyclic diester.

[1]

(ii) Poly(lactic acid), PLA, is used to make 'dissolvable' stitches (for holding wounds together). PLA breaks down into smaller molecules after one or two weeks.

Draw the structure of **one** repeat unit in PLA.

[1]

(iii) Explain how PLA breaks down and why the stitches 'dissolve'.

....

....

....

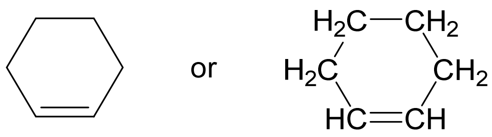
....

....[2]

[Total: 13]

For Examiner's Use

5 Cyclohexene behaves as a typical alkene.



(a) **(i)** Give the name of the type of polymerisation that cyclohexene undergoes.

....[1]

(ii) Draw a section of the polymer consisting of **three** repeat units.

[1]

(iii) A sample of cyclohexene is polymerised to a relative molecular mass of 2500, on average.

Calculate the number of complete cyclohexene units that polymerise in each polymer molecule on average. Show your working.

number of cyclohexene units = ... [1]

(b) Cyclohexene will react with hydrogen.

(i) Write an equation for the reaction.

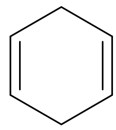
[1]

(ii) State the conditions required.

....[1]

For Examiner's Use

(c) Cyclohexa-1,4-diene also displays reactivity typical of alkenes. Its structure is shown.



Draw the structures of all possible products of the reaction when one molecule of cyclohexa-1,4-diene completely reacts with two molecules of **hydrogen bromide**.

[2]

(d) A graphene sheet is a layer of graphite.

A recent development has been the synthesis of graphene ribbons (reported in Nature, 2010). A reaction scheme is shown Fig. 5.1.

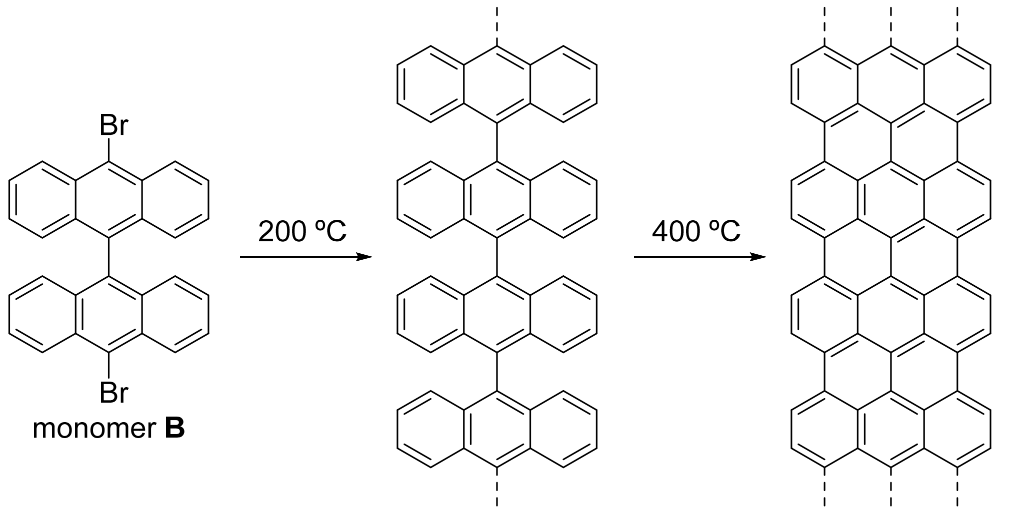


Fig. 5.1

(i) When monomer **B** is polymerised to make **C** there is also another product, **X**.

Give the molecular formula of **X**.

X is .... [1]

For Examiner's Use

(ii) In the transformation of polymer **C** into polymer **D**, another product, **Y,** is produced.

Give the molecular formula of **Y**.

Y is .... [1]

(iii) Deduce the number of moles of **X** and **Y** produced per mole of monomer **B**.

number of moles of **X** = ...

number of moles of **Y** = ... [2]

(iv) Boron nitride, BN, forms sheets similar to graphene except they contain dative covalent bonds as well as covalent bonds.

Add all the possible **dative** covalent bonds between the atoms shown in the structure Fig. 5.2.

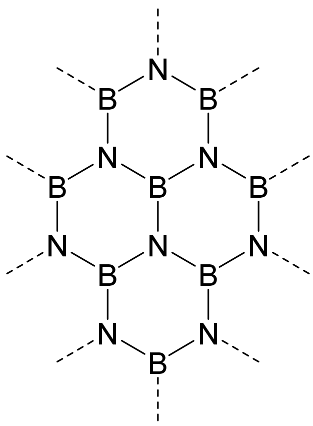


Fig. 5.2

[1]

(v) Boron nitride can also form a giant covalent structure in which each atom has four single bonds.

Suggest the name of another substance which has this type of structure.

....[1]

[Total: 13]

For Examiner's Use

Section B

Answer **one** question in this section, in the space provided.

6 **(a)** Under certain conditions the halogen halides decompose into their elements.

$$2HX \longrightarrow H_2 + X_2 \quad (X = Cl, Br, I)$$

(i) How can this reaction be carried out?

....[1]

(ii) Describe what will be observed in the case of HI.

....

....[1]

(iii) Use bond energy values from the *Data Booklet* to calculate ΔH for this reaction when X = C*l* and when X = I.

[2]

(iv) Explain how the extent of decomposition of the hydrogen halides vary down Group 17 using the results from **(a)(iii)**.

....

....

....

....

....[2]

For Examiner's Use

(b) The dissociation of water is a reversible reaction.

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

The ionic product of water, K_w, measures the extent of dissociation of water.

K_w varies with temperature. Therefore, it is always important to quote the temperature at which measurements are being taken.

Fig. 6.1 shows the variation of K_w between 0 ºC and 60 ºC.

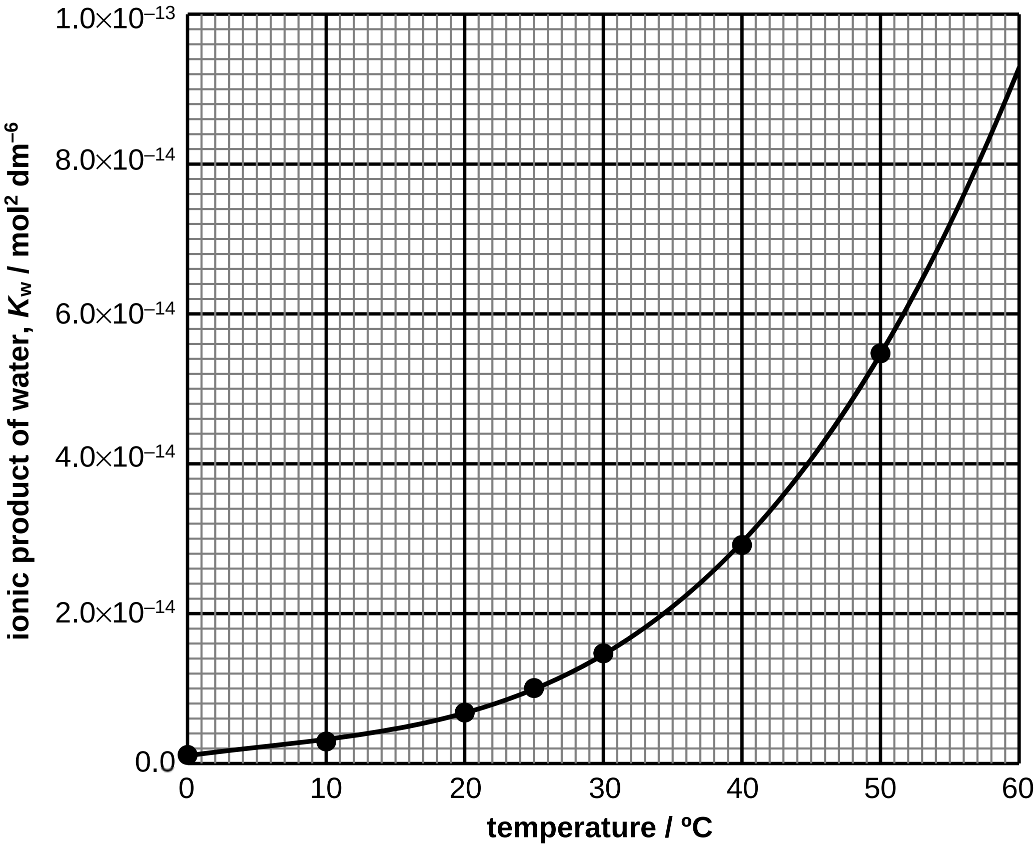


Fig. 6.1

(i) Write the expression for K_w.

....[1]

(ii) Calculate the OH^-(aq) concentration in an aqueous solution of hydrochloric acid with a pH of 4.37 at 25 ºC.

[2]

For Examiner's Use

(iii) Using Fig. 6.1, explain whether the dissociation of water is an exothermic or endothermic process.

....

....

....[1]

(iv) Determine the pH of pure water at body temperature, 37 ºC.

[3]

(c) In humans it is important for blood to be maintained at a pH between 7.35 and 7.45. One of the way it does this is by using a buffer of CO_2(aq) and HCO_3^- (aq).

During vigorous exercise the muscles produce lactic acid. Lactic acid is transported by the blood to the liver to be broken down.

(i) State the effect that exercise will have on the pH of blood if there is no buffer present.

....[1]

(ii) Write equation(s) to explain how the CO_2(aq) / HCO_3^- (aq) buffer system helps to maintain the pH of blood between 7.35 and 7.45 during exercise.

....

....

....

....

....[2]

For Examiner's Use

(c) Chlorine reacts with both propane and propene, but under different conditions.

Write equations, giving the structural formulae of the organic products, and state the conditions for the reaction of chlorine.

(i) with propane

....[2]

(ii) with propene

....[2]

[Total: 20]

For Examiner's Use

7 (a) In the presence of acid, $H^+(aq)$, aqueous bromate(V) ions, $BrO_3^-(aq)$, react with aqueous bromide ions, $Br^-(aq)$, to produce bromine, $Br_2(aq)$.

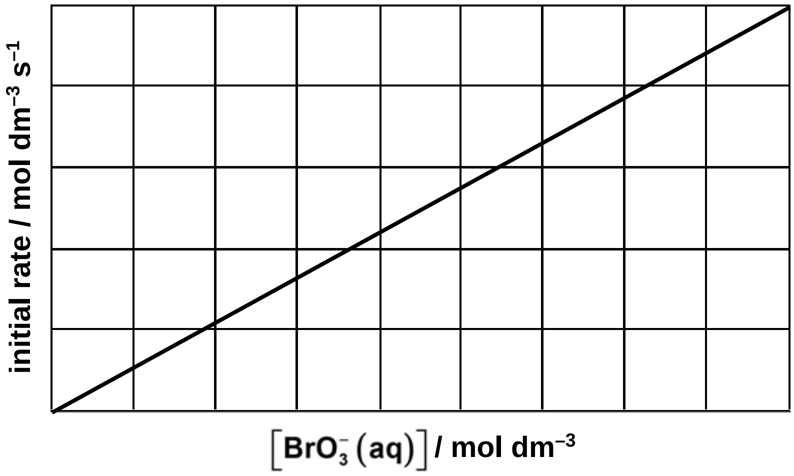
A student carried out an investigation into the kinetics of this reaction.

(i) Balance the ionic equation for this reaction.

BrO_3^- + ………. Br^- + ………. H^+ → ………. Br_2 + …….. H_2O [1]

The student investigated how different concentrations of $BrO_3^-(aq)$ affect the initial rate of the reaction.

A graph of initial rate against $\left[BrO_3^-(aq)\right]$ is shown below.



The student then investigated how different concentrations of $Br^-(aq)$ and $H^+(aq)$ affect the initial rate of the reaction.

The results are shown below.

$\left[BrO_3^-(aq)\right]$ / mol dm^{-3}	$\left[Br^-(aq)\right]$ / mol dm^{-3}	$\left[H^+(aq)\right]$ / mol dm^{-3}	initial rate / mol dm^{-3} s^{-1}
5.0×10^{-2}	1.5×10^{-1}	3.1×10^{-1}	1.19×10^{-5}
5.0×10^{-2}	3.0×10^{-1}	3.1×10^{-1}	2.38×10^{-5}
5.0×10^{-2}	1.5×10^{-1}	6.2×10^{-1}	4.76×10^{-5}

For Examiner's Use

(ii) Using the results from the student's experiments, determine the order of reactions with respect to all the three reactants.

order w.r.t. BrO_3^- = ...

order w.r.t. Br^- = ...

order w.r.t. H^+ = ...[3]

(iii) Write the rate equation and calculate the rate constant for this reaction, including the units.

[3]

(b) Sodium, magnesium, aluminium, silicon, phosphorus, sulfur, chlorine and neon are elements in period 3 of the periodic table.

(i) State and explain the differences between the ionic radii of Na^+, Si^{4+}, and P^{3-} and Cl^-.

....

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....

....

....

....

....[2]

For Examiner's Use

(ii) Sodium, silicon, phosphorus and chlorine also differ greatly in their melting points. Describe how their melting points differ, and explain this variation.

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....

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....

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....

....[4]

(iii) The acid-base behaviour of aluminium oxide, Al_2O_3, shows similarities to that of magnesium oxide, MgO, on the one hand, and phosphorus(V) oxide, P_4O_{10}, on the other.

Describe what these similarities are, and explain why aluminium oxide occupies this in-between position.

Write equations for all the reactions you choose to illustrate your answer.

....

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....[5]

For Examiner's Use

(c) Sodium borohydride, $NaBH_4$, is a useful reducing agent in organic chemistry. It reduces both propanal and propanone.

Write balanced equations, giving the structural formulae of the organic products, to illustrate its reducing property.

(i) with propanal

....[1]

(ii) with propanone

....[1]

[Total: 20]